D 30586 **(Pages : 2)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION, NOVEMBER 2022

Physics/Applied Physics

PHY 5B 08/APH 5B 08—OPTICS

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A (Short Answer Type)

Answer **all** *questions in two* **or** *three sentences.*
Each correct answer carries a maximum of 2 *marks.*

1. State the laws of reflection.
2. Explain the law of cosine square for the intensity distribution in the interference of two waves of the same amplitude.
3. List any four conditions for obtaining a distinct well-defined interference pattern.
4. Explain the origin for the colours of thin films.
5. Give an expression for the intensity distribution in Fraunhofer diffraction due to N slits and explain the terms involved.
6. Give the grating equation and explain the terms involved.
7. What do you mean by the resolving power of a grating ? Give an expression for the same explaining the terms involved.
8. Give a figure illustrating the Huygens wave surfaces produced by a point source embedded in a negative doubly refracting crystal.
9. What do you mean by the optical activity of certain substances ? Give two examples of optically active substances.
10. Mention the applications of holography.
11. What do you mean by a graded index fiber ? Draw its refractive index profile.
12. What do you mean by pulse dispersion in optical fiber ?

(Ceiling - 20 marks)

Turn over

Section B (Paragraph/Problem Type)

Answer **all** *questions in a paragraph of about* **half a page to one page.**
Each correct answer carries a maximum of 5 *marks.*

13. What do you mean by a thin lens? Write down the thin lens formula and explain the terms involved.

14. Interference fringes are observed with a biprism of refracting angle 1° and refractive index 1.5 on a screen 0.8 m from it. If the distance between the source and the biprism is 0.2 m, estimate the fringe width when a light of wavelength 690 nm is used.

15. Obtain the radii of the first two dark rings of the Fraunhofer diffraction pattern produced by a circular aperture of radius 0.02 cm at the focal plane of a convex lens of focal length 20 cm. Given, the wavelength λ = 600 nm.

16. Consider a diffraction grating having 15000 lines per inch. Obtain the angular separation of the D_1 and D_2 lines of sodium in the second order spectra.

17. Determine the radius of the first zone in a zone plate of focal length 20 cm for light of wavelength 500 nm.

18. Using a suitable figure, explain the reconstruction of image in holography.

19. A step index fiber has a core of refractive index 1.55 and cladding of refractive index 1.5. Determine the numerical aperture of the fiber. Assume that light enters the fiber from air.

(Ceiling - 30 marks)

Section C (Essay Type)

Answer in about **two** *pages, any* **one** *question.*
Correct answer carries 10 *marks.*

20. Using a suitable figure, discuss the principle, construction and working of a Michelson interferometer. Discuss the condition for obtaining brightness in circular rings.

21. Using suitable figures, explain the optical activity of substances. Discuss the Fresnel's explanation of optical rotation.

(1 × 10 = 10 marks)